

Personalized Education Agent

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Introduction



Education struggles to scale personalized learning



Most platforms follow static, one-size-fits-all paths



Students learn at different paces, with unique gaps and needs



We propose the AI-powered Personalized Education Agent to adapt in real time

Importance

Uses an agentic approach driven by interaction data

Predicts learner mastery dynamically

Generative AI engine creates content at the right difficulty level

Provides timely, meaningful feedback for every learner

Responds directly to the growing demand for adaptive learning solutions

Goal



Deliver a working prototype of the Personalized Education Agent



Create a system that adapts to each student individually



Improve engagement, motivation, and knowledge retention



Advance innovation in the field of intelligent tutoring systems

Issues / Challenges

01

Designing accurate student modeling & knowledge tracing

02

Generating adaptive content that balances challenge and support

03

Providing real-time feedback that guides and motivates

04

Building educator dashboards that are simple yet powerful

05

Ensuring effectiveness through small-scale usability testing

Methodology



- **Student Modeling & Tracing:** Track progress, errors, and learning patterns
- **Adaptive Content Generation:** AI creates personalized questions & explanations
- **Real-Time Feedback:** Offer instant guidance, corrections, and encouragement
- **Educator Dashboards:** Show class-wide and individual progress at a glance
- **Usability Testing:** Pilot with learners and teachers for refinement

Expected Outcomes



Knowledge Tracker → estimates student understanding, predicts next step



Content Creator → generates relevant and customized practice content



Feedback Engine → provides instant, supportive explanations



Teacher Dashboard → highlights progress, gaps, and engagement trends



Success measured by **accuracy, usability, and improved student outcomes**